

# Shihao Weng

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## Education

- 2026.10 – Present 📖 **Visiting Ph.D. Student**, The Chinese University of Hong Kong, Hong Kong SAR.  
Advisor: Prof. Michael R. Lyu.
- 2025.09 – 2026.09 📖 **Visiting Ph.D. Student**, Singapore Management University, Singapore.  
Advisor: Prof. Xiaofei Xie.
- 2024.09 – Present 📖 **Ph.D. in Software Engineering**, Nanjing University, China.  
Advisor: Prof. Yang Feng.
- 2022.09 – 2024.06 📖 **M.S. in Software Engineering**, Nanjing University, China.
- 2018.09 – 2022.06 📖 **B.E. in Computer Science and Technology**, Jiangnan University, China.

## Research Publications

- 1 **Shihao Weng**, Yang Feng, Xiaofei Xie, and Jiongchi Yu. “AgentAntibody: An Adaptive Immune System for Defending LLM Agents against Prompt Injection.” In: arXiv Preprint (arXiv: 2608.04053, 2026).
- 2 **Shihao Weng**, Yang Feng, and Xiaofei Xie. “Beyond Accuracy: Policy Invariance as a Reliability Test for LLM Safety Judges.” In: arXiv Preprint (arXiv: 2605.06161, 2026).
- 3 **Shihao Weng**, Yang Feng, Jinrui Zhang, Xiaofei Xie, Jiongchi Yu, and Jia Liu. “ARGUS: Defending LLM Agents Against Context-Aware Prompt Injection.” In: arXiv Preprint (arXiv: 2605.03378, 2026).
- 4 **Shihao Weng**, Yang Feng, Jincheng Li, Yining Yin, Zhenlun Zhang, Lyuxi Liu, and Jia Liu. “TACO: Trust Assessment of Large Language Models in Coding Assistance Tasks.” In: *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (ICSE)*, CCF-A.
- 5 Zhenlun Zhang, Yang Feng, **Shihao Weng**, Yining Yin, Jincheng Li, and Jia Liu. “VADA: A Multicultural Benchmark for Value-Aware Data Generation and Alignment Evaluation in LLMs.” In: *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (ICSE)*, CCF-A.
- 6 **Shihao Weng**, Yang Feng, Jincheng Li, Yining Yin, Xiaofei Xie, and Jia Liu. “AtPatch: Debugging Transformers via Hot-Fixing Over-Attention.” In: *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (ICSE)*, CCF-A.
- 7 **Shihao Weng**, Yang Feng, Yining Yin, Zhenlun Zhang, and Baowen Xu. “Data Preparation and Quality for Code-Centric Generative Software Engineering Tasks: A Systematic Literature Review.” In: *Frontiers of Computer Science*, CCF-B.
- 8 Yining Yin, Yang Feng, **Shihao Weng**, Xinyu Gao, Jia Liu, and Zhihong Zhao. “Lightweight Probabilistic Coverage Metrics for Efficient Testing of Deep Neural Networks.” In: *Proceedings of the 16th Asia-Pacific Symposium on Internetware*, CCF-C.
- 9 **Shihao Weng**, Yang Feng, Yining Yin, Yuxuan Dai, Jia Liu, and Zhihong Zhao. “Seeing the Invisible: Test Prioritization for Object Detection System.” In: *Empirical Software Engineering (EMSE)*, CCF-B.
- 10 Yining Yin, Yang Feng, **Shihao Weng**, Yuan Yao, Jia Liu, and Zhihong Zhao. “Dataactive: Data Fault Localization for Object Detection Systems.” In: *Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, CCF-A.

- 11 Yining Yin, Yang Feng, **Shihao Weng**, Zixi Liu, Yuan Yao, Yichi Zhang, Zhihong Zhao, and Zhenyu Chen. “Dynamic Data Fault Localization for Deep Neural Networks.” In: *Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (FSE)*, CCF-A.
- 12 **Shihao Weng**, Yang Feng, Yining Yin, and Jia Liu. “Prioritizing Testing Instances to Enhance the Robustness of Object Detection Systems.” In: *Proceedings of the 14th Asia-Pacific Symposium on Internetware*, CCF-C; Distinguished Paper Award.

## Honors and Academic Service

### Honors and Awards

- 2023–2025  **Nanjing University Outstanding Postgraduate Student Award** (three consecutive years).
- 2023  **National Scholarship for Graduate Students**, awarded by the Ministry of Education of China.
- 2021  **CCF Outstanding Undergraduate Student**, awarded by the China Computer Federation.
-  **Bronze Medal**, 45th ACM International Collegiate Programming Contest (ICPC), Asia Regional Contest.
- 2020–2021  **National Scholarship for Undergraduate Students** (two consecutive years), awarded by the Ministry of Education of China.

### Academic Service

- 2026  **Reviewer**, NeurIPS 2026 Workshop (TAE).